

Power BI

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Task3



Introduction to Power BI

What is Power BI & Why Use It?

Power BI is a **data visualization and business intelligence tool by Microsoft**. It helps you **turn raw data into interactive dashboards and reports** — so you can make better business decisions.

Why use Power BI?

- Easy to connect with multiple data sources (Excel, SQL, Web, etc.)
- Converts data into beautiful visuals and dashboards
- Real-time insights for decision-making
- Cloud sharing and collaboration
- Regular updates from Microsoft

Power BI Desktop vs Power BI Service vs Power BI Mobile

Version	Description	Used For
Power BI Desktop	Free Windows application	Used to create reports and dashboards . You do all your data cleaning, modeling, and visualization here.
Power BI Service (Cloud)	Online version (app.powerbi.com)	Used to publish, share, and collaborate on reports. Dashboards can be refreshed automatically and shared with teams.

Version	Description	Used For
Power BI Mobile	App for Android/iOS	Used to view and interact with dashboards and reports on the go.

Power BI Workflow

Power BI follows a simple 4-step process:

Connect → Transform → Visualize → Share

1. **Connect** – Link Power BI to various data sources (Excel, databases, APIs, etc.)
 2. **Transform** – Clean, filter, and shape data using Power Query Editor
 3. **Visualize** – Create charts, graphs, and dashboards using drag-and-drop tools
 4. **Share** – Publish and share insights via Power BI Service or Mobile app
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Installation & Setup

Steps:

1. Go to <https://powerbi.microsoft.com/desktop>
 2. Download **Power BI Desktop (Free)**
 3. Install it like any Windows software (Next → Next → Finish)
 4. Open it and **sign in with your Microsoft account**
 5. Start by clicking “**Get Data**” to import your first dataset
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Getting Data (Data Sources)

Power BI can pull data from **many different places** — files, databases, web sources, and more.

This is the **first step** in building any Power BI report.

Importing from Files

Power BI supports many file types:

File Type	Description	How to Import
Excel (.xlsx)	Most common data source with sheets and tables	Home → Get Data → Excel → Select file
CSV (.csv)	Plain text file with comma-separated values	Home → Get Data → Text/CSV → Load file
Text (.txt)	Data stored in lines or tab-delimited format	Home → Get Data → Text/CSV
JSON (.json)	Used in APIs or web data formats	Home → Get Data → JSON

➔ *Power BI automatically detects columns, headers, and datatypes when importing.*

Connecting to Databases

You can connect Power BI directly to popular databases to fetch live or stored data.

Database	Connection Method
SQL Server	Get Data → SQL Server → Enter Server Name & Database → Load
MySQL	Get Data → Database → MySQL Database → Enter credentials
PostgreSQL	Get Data → Database → PostgreSQL → Enter credentials
Oracle	Get Data → Database → Oracle Database → Enter connection info

➔ *Use this when your organization stores data centrally in a database instead of Excel.*

Web Data Import (APIs, Web Pages, Web Scraping)

Power BI can also fetch data from the internet:

- **From a webpage:**
Home → Get Data → **Web** → Paste the URL
→ Power BI detects tables on that page → Load
- **From APIs:**
Use API URL (like <https://api.example.com/data>)
→ often requires **API Key** or **token**
- **For Web Scraping:**
You can extract tabular data (like product lists or stock prices) from websites.

➔ *Useful for pulling data that updates online regularly (e.g., weather, stock, or news feeds).*

DirectQuery vs Import Mode

Mode	Description	Pros	Cons
Import Mode	Data is copied into Power BI	Fast performance, offline access	Must refresh data manually
DirectQuery	Connects live to database	Always up-to-date data	Slower performance, needs internet connection

- ➔ *Use Import for smaller datasets; use DirectQuery for large, frequently changing data.*

Connecting Multiple Sources Together

Power BI allows you to **combine different data sources** in one report:

- Example: Join **Excel sales data** with **SQL customer data**
 - Use **Power Query Editor** → **Merge Queries** / **Append Queries**
 - Create **relationships** between tables (like CustomerID, ProductID)
- ➔ *This helps build a 360° dashboard combining info from multiple departments or systems.*

3 Power Query Editor (Data Cleaning & Transformation)

The **Power Query Editor** in Power BI is the place where you **clean, correct, and prepare data** before using it in reports.

Think of it like an “Excel Power Tool” — it helps you **fix messy data** so your visuals are accurate.

1. Removing Nulls & Duplicates

- **Nulls:** These are blank or missing values.
 - Use **Home** → **Remove Rows** → **Remove Blank Rows** or **Replace Values** → **Replace null with 0 / N/A**
 - **Duplicates:** These are repeated entries.
 - Select a column → **Remove Rows** → **Remove Duplicates**
- ➔ *This keeps your data clean and reliable.*
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2. Filtering Rows & Columns

- **Filter Rows:** Keep only the rows you want.
Example: Keep only records where “Sales > 5000.”
 - **Remove Columns:** Delete unnecessary columns that you don’t need for analysis.
 - **Home** → **Remove Columns**
- ➔ *Filtering reduces file size and focuses on useful data.*
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3. Data Type Conversion

Each column should have a **correct data type** like:

- Text
- Whole Number
- Decimal Number
- Date/Time

► **Transform → Data Type → Choose Type**

➔ *Correct data types are important for accurate calculations and sorting.*

4. Splitting & Merging Columns

- **Split Column:** Break a column into parts.
Example: Split “Full Name” → “First Name” + “Last Name”
► **Split Column → By Delimiter**
 - **Merge Columns:** Combine two or more columns.
Example: “City” + “State” → “City, State”
► **Add Column → Merge Columns**
- ➔ *Helps you reshape data as needed.*
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5. Replace Values

Use this to fix or standardize entries.

Example: Replace “N/A” with “Not Available,” or “Male/Female” → “M/F.”

► **Transform → Replace Values**

➔ *Makes data consistent and readable.*

6. Append & Merge Queries

- **Append Queries:** Stack tables vertically (like combining months’ data).
► Jan + Feb + Mar → One full dataset

- **Merge Queries:** Join tables side by side using a key column (like in SQL joins).
 - Join “Sales Table” with “Customer Table” using “CustomerID.”
- ➔ *Used when your data is spread across multiple tables.*
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7. Creating Calculated Columns

Add new columns based on formulas.

Example:

Total = [Quantity] * [Price]

➤ **Add Column → Custom Column**

➔ *Used to create new insights from existing data.*

8. Conditional Columns

Make rule-based columns (like IF conditions).

Example:

If Sales > 10000 → “High”, else “Low.”

➤ **Add Column → Conditional Column**

➔ *Helps categorize data automatically.*

9. Date & Time Transformation

Power BI can break down and calculate date/time easily:

- Extract **Year, Month, Day, Quarter, Weekday**
- Create **Duration** between two dates
 - **Add Column → Date / Time Menu**

➔ *Useful for time-based analysis and trends.*

10. Best Practices for Data Modeling

Rename columns clearly

Keep only necessary data

Ensure correct data types

Maintain clean relationships between tables

Refresh data regularly

Always preview changes before loading

➔ *Good modeling = faster reports + accurate results.*

Data Modeling in Power BI

- Data modeling is the **heart of Power BI** — it defines how your **tables connect and interact** with each other.
 - A good model helps you analyze data faster, avoid errors, and build flexible dashboards.
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1. Star Schema vs Snowflake Schema

Star Schema

- Central **Fact Table** (contains numbers like Sales, Revenue, Quantity)
- Connected to multiple **Dimension Tables** (contains descriptive info like Customer, Product, Region)
- All dimensions directly connect to the fact table.

Example:

Sales Table (Fact)

|

├── Customer Table

├── Product Table

├── Region Table

└── Date Table

➔ *Simple, fast, and best for Power BI.*

Snowflake Schema

- Like a star schema, but **dimension tables are further split** into sub-dimensions.

Example: Product → Category → Department

✓ *More normalized (less data repetition),*

✗ *but slightly slower and complex for Power BI visuals.*

➔ **Tip:** Power BI performs best with **Star Schema** — it's easier to understand and faster to calculate.

2. Relationships Between Tables

Relationships define how tables **connect through key fields**.

Type	Meaning	Example
One-to-Many (1:*):	One record in Table A connects to many in Table B	One Customer → Many Orders
Many-to-Many (:):	Both tables can have multiple matching records	Students & Courses
One-to-One (1:1):	One record matches exactly one record	Employee & EmployeeDetails

➔ *Power BI automatically detects relationships when you load data, but you can edit them manually.*

3. Active & Inactive Relationships

Sometimes two tables can be connected by **multiple relationships**, but **only one can be active** at a time.

Example:

- Sales table connects to Date table by **OrderDate** and **ShipDate**.
 - Only **OrderDate** might be active.
 - You can activate **ShipDate** when needed using DAX:
- `CALCULATE([Total Sales], USERRELATIONSHIP(Sales[ShipDate], Date[Date]))`

✓ *Active = used by default in visuals*

✓ *Inactive = can be used manually in formulas.*

4. Managing Relationship Direction (Single vs Both)

Direction	Flow of Filter	Use Case
Single Direction	Filter flows from one table to another (default)	e.g., Customer → Sales
Both Directions	Filter flows both ways	For many-to-many relationships or complex filters

➔ *Use “Both” only when necessary — it can slow performance or cause ambiguity.*

5. Role-Playing Dimensions (Multiple Date Fields)

Some fact tables have **multiple date fields** like:

- OrderDate
- ShipDate
- DeliveryDate

You can't use one date table for all directly.

✓ Solution → **Duplicate the Date Table** for each role:

- Date_Order for Order Date

- Date_Ship for Ship Date

➔ *This allows accurate time-based analysis across different date types.*

Summary

Concept	Meaning
Star Schema	Simple model with one fact table and related dimensions
Snowflake Schema	More complex model with sub-dimensions
Relationships	Define how tables connect
Active/Inactive	Control which relationship is in use
Relationship Direction	Controls filter flow
Role-Playing Dimensions	Use duplicate date tables for multiple date fields

Data Visualization (Reports & Dashboards)

Data visualization in Power BI is all about **turning data into stories** — using charts, maps, and dashboards that make insights easy to understand and act on.

A good Power BI report should be **clear, simple, and interactive**.

1. Report Design Principles (Clarity, Simplicity)

When designing Power BI reports:

- **Clarity:** Use clean layouts and clear titles
- **Simplicity:** Avoid too many visuals or colors

- **Consistency:** Same color theme, fonts, and chart style
- **Focus:** Highlight key insights (KPIs, trends, comparisons)
- **Interactive:** Allow users to explore data with filters or drill-downs

➔ *Remember: A great dashboard tells a story in one glance.*

2. Built-in Visuals in Power BI

Power BI offers many **default visuals** you can drag and drop easily:

Visual Type	Purpose / Example
Bar / Column Chart	Compare values (e.g., Sales by Region)
Line Chart	Show trends over time (e.g., Monthly Sales)
Area Chart	Show trends with filled color under line
Pie / Donut Chart	Show proportion or percentage (e.g., Market Share)
Map / Filled Map	Show data by geographic location
Cards	Display single numbers (e.g., Total Sales, Profit)
Tables / Matrix	Show detailed, structured data like Excel
Gauge / KPI	Show progress toward a goal or target

➔ *Combine visuals smartly — too many charts can confuse viewers.*

3. Custom Visuals from Marketplace

If built-in visuals aren't enough:

- Go to **Visualizations Pane** → **Get More Visuals** → **AppSource**
- Download from **Power BI Marketplace**

Examples:

Funnel charts, Word clouds, Bullet charts, Radar charts, Advanced KPI visuals, etc.

➔ *Use custom visuals to make your dashboard more engaging.*

4. Conditional Formatting

Conditional formatting helps highlight important data automatically. You can apply colors, icons, or data bars based on values.

Example:

- Green = High sales
 - Red = Low sales
- **Format** → **Conditional Formatting** → **Choose rule**
- ➔ *Visually emphasizes good or bad performance.*
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5. Drill-Downs & Drill-Throughs

Feature	Description	Example
Drill-Down	View data at deeper levels within the same visual	From Year → Quarter → Month
Drill-Through	Click on a data point to open another page with detailed info	Click on “Product A” → opens “Product Details” page

➔ *Great for storytelling — users can explore data in depth.*

6. Tooltips & Report Page Tooltips

- **Tooltips:** Small info boxes that appear when you hover over a data point
Example → Hover on bar → shows Sales, Profit, and Region
- **Report Page Tooltips:** Custom-designed tooltip pages with extra visuals or context

➔ *Adds hidden insights without cluttering the main page.*

7. Bookmarks & Selections (Interactive Reports)

- **Bookmarks:** Save a specific view of your report (filters, visuals, etc.)
Example → “Top 5 Products View”
 - **Selections:** Control which visuals are visible for each bookmark
➔ *Used to create buttons for navigation (like mini dashboards or tabs).*
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8. Hierarchies (Date, Product Categories, Regions)

Hierarchies let users drill down from **summary** → **detail** levels.

Examples:

- **Date Hierarchy:** Year → Quarter → Month → Day
 - **Product Hierarchy:** Category → Subcategory → Product
 - **Region Hierarchy:** Country → State → City
- ➔ *Makes navigation easier and helps view trends at different levels.*
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9. Slicers & Filters

Filters control what data is visible — they can be applied at different levels.

Level	Description	Example
Visual-level	Affects only one chart	Filter one bar chart by region
Page-level	Affects all visuals on one page	Show only “2024” data
Report-level	Affects all pages	Show only “India” sales

Slicers:

Visual filters users can click (like buttons).

Example: Slicer for **Year**, **Category**, or **Region** → lets users interactively change data view.

➔ *Always give users control through slicers for flexible analysis.*

Summary

Concept	Key Idea
Design Principles	Keep it clean, clear, and focused
Built-in Visuals	Bar, Line, Pie, Map, Cards, Tables
Custom Visuals	Download extra visuals from AppSource
Conditional Formatting	Highlight data based on values
Drill-Down / Drill-Through	Explore deeper details
Tooltips	Show extra info on hover
Bookmarks	Save specific views for interactivity
Hierarchies	Organize data (e.g., Year → Month)
Slicers & Filters	Let users control data visibility

Advanced Features in Power BI

Once you know the basics, Power BI offers advanced tools to make your reports more **dynamic, powerful, and user-friendly**.

Let's go through them one by one 

1. Parameters & What-If Analysis

Parameters let you create **dynamic inputs** — users can change values and instantly see results.

Used for **scenarios, sensitivity tests, or target analysis**.

Example:

You can create a “*Discount %*” *parameter* → change it using a slider → and see how profit changes.

✦ Steps:

1. Go to **Modeling** → **New Parameter** → **Numeric Range**
 2. Set min, max, and increment values
 3. Power BI automatically creates a slicer for users to interact with
➔ *What-If analysis helps test different business scenarios interactively.*
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2. Grouping & Binning Data

Used to **organize or simplify** your data visually.

- **Grouping:** Combine similar categories together
→ Example: Group small regions into “Others”
- **Binning:** Divide continuous numeric data into ranges (bins)
→ Example: Group customers by age ranges (18–25, 26–35, etc.)

✦ Steps:

Right-click a field → **New Group** → Choose grouping or binning options.

➔ *Helps make charts clearer and comparisons easier.*

3. Creating Hierarchies

Hierarchies create **drill-down paths** in visuals.

Example hierarchies:

- **Date:** Year → Quarter → Month → Day
- **Product:** Category → Subcategory → Product
- **Region:** Country → State → City

✦ Steps:

In **Fields Pane**, drag a field onto another to create a hierarchy.

➔ *Makes navigation and drill-down seamless for users.*

4. Sync Slicers Across Pages

If your report has multiple pages, you can **sync slicers** so users don't need to reselect filters again and again.

✦ Steps:

1. Add slicer on Page 1
 2. Go to **View** → **Sync Slicers Pane**
 3. Select pages to sync it with (e.g., Page 2, Page 3)
- ➔ *Ensures consistent filtering experience across all report pages.*

5. Row-Level Security (RLS)

RLS controls **who can see what data** — ensuring data privacy.

Example:

- Sales Manager A → can only see “North Region” sales
- Sales Manager B → can only see “South Region” sales

✦ Steps:

1. Go to **Modeling** → **Manage Roles**
 2. Define rules like:
Region[Name] = "North"
 3. Assign users to roles in **Power BI Service**
- ➔ *Essential for corporate dashboards with multiple users.*
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6. Q&A (Natural Language Query)

Power BI allows users to **type questions in plain English** and get instant visual answers.

Example:

“Show total sales by region for 2024”

Power BI automatically creates a chart!

✨ Steps:

- Add a **Q&A visual**
 - Users can ask questions using natural language
 - ➔ *Great for non-technical users to explore data easily.*
-

7. Power BI Apps & Templates

Power BI Apps

- Collections of related dashboards and reports
- Can be shared easily with teams or departments
- Example: “Sales Performance App,” “Finance Dashboard App”

✨ Steps:

Publish report → Create app → Share via Power BI Service

Templates

- Pre-built report structures with data models and visuals
 - Useful for standardizing reporting across teams
 - ➔ *Apps = packaged reports; Templates = reusable report blueprints.*
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Summary

Feature	What It Does	Example
Parameters / What-If	Create dynamic, adjustable values	Test different discount rates
Grouping / Binning	Combine or categorize data	Group small regions
Hierarchies	Enable drill-down navigation	Year → Quarter → Month
Sync Slicers	Keep slicers consistent across pages	Same “Year” filter everywhere
Row-Level Security	Restrict data visibility	North vs South manager
Q&A	Ask questions in natural language	“Top 5 products by sales”
Apps & Templates	Package or reuse reports	Department dashboards

8 Power BI Service (Cloud & Collaboration)

After creating reports in **Power BI Desktop**, the next step is to **publish and share them** online using **Power BI Service** — Microsoft’s **cloud platform** for Power BI.

It’s where you **collaborate, refresh data, and manage access** securely.

1. Publishing Reports from Desktop to Service

Once your report is ready in **Power BI Desktop**, you can **publish it to the cloud** (Power BI Service) to share it with others.

 Steps:

1. Sign in to your Microsoft Power BI account in Desktop.
2. Click **Publish** → **Select Workspace** → **Done!**

 The report and its dataset appear in your Power BI Service workspace.

➔ *Publishing = moving your local report to the online Power BI platform.*

2. Workspaces & Apps

Power BI Service organizes reports into **Workspaces** and **Apps**.

◆ Workspaces

- Shared environments for teams to **collaborate** on reports and datasets.
- Example: “Sales Workspace,” “Marketing Workspace.”
- You can **assign roles** (Admin, Member, Contributor, Viewer).

◆ Apps

- Packaged versions of workspaces for easy **view-only access**.
- Example: Create a “Finance Dashboard App” → share with all finance users.

➔ *Workspaces = for creating & managing; Apps = for sharing with audience.*

3. Sharing Reports & Dashboards

In Power BI Service, you can **share** reports with individuals or groups.

🌟 Options:

- **Share directly** via email or link (with permissions)
- **Publish to web** (public, for non-sensitive content)
- **Embed** in websites or Teams
- **Collaborate in Microsoft Teams** integration

➔ *Sharing ensures everyone can view the latest version of your report, anytime.*

4. Data Refresh (Manual & Scheduled)

Your reports depend on data that may change frequently.

Power BI Service lets you **refresh** datasets to keep them updated.

- **Manual Refresh**

Click **Refresh Now** in the dataset menu.

- **Scheduled Refresh**

Set automatic refresh (e.g., daily at 8 AM).

➔ *Pro Tip:* You need a **gateway** for on-premises data sources.

5. Data Gateways (Personal & Enterprise)

A **Data Gateway** connects Power BI Service (cloud) to your local or on-premise data (like SQL Server or Excel on your PC).

Types:

- **Personal Gateway:** For individual use; installed on your PC.
- **Enterprise Gateway:** For teams; centrally managed by IT.

➔ *Gateways ensure data refresh works securely between your local files and the cloud.*

6. Usage Metrics (Report Usage Tracking)

Power BI automatically tracks **how users interact** with your reports.

You can view:

- Number of views
- Who viewed it
- Which visuals are most used
- Time spent on each page

🔧 Steps:

In Power BI Service → Go to report → **View Usage Metrics Report**

➔ *Helps you understand audience engagement and improve report design.*

7. Collaboration & Commenting Features

Power BI Service supports **real-time collaboration**:

- **Comments:** Users can tag others (e.g., @Kriti) to discuss visuals.
 - **Sharing links:** Collaborators can co-edit or explore together.
 - **Integration with Teams & SharePoint:** Discuss reports directly in chat.
- ➔ *Makes Power BI a team tool — not just a reporting tool.*

Summary

Feature	What It Does	Example
Publish Reports	Move from Desktop → Cloud	Share dashboards online
Workspaces & Apps	Organize & share content	“Marketing Workspace” app
Sharing Options	Give others access	Share via Teams or email
Data Refresh	Keep reports updated	Daily refresh at 8 AM
Data Gateways	Connect local → Cloud	SQL Server → Power BI
Usage Metrics	Track viewer activity	500 users viewed dashboard
Collaboration & Comments	Work together	@tag teammates in visuals

In short:

Power BI Service is where your hard work in Power BI Desktop turns into a **living, collaborative dashboard** — always fresh, shared, and accessible from anywhere

Power BI with Other Tools

Power BI doesn't work in isolation — it connects seamlessly with **Microsoft tools, databases, and web platforms**, allowing users to **analyze, share, and**

embed insights across systems.

Let's understand each integration 🖐️

1. Power BI + Excel Integration (Analyze in Excel)

Power BI and Excel are like best friends — both are used for data analysis but at different levels.

Excel is great for **manual calculations**, while Power BI handles **automation and visuals**.

◆ Integration Features:

- **Analyze in Excel:**

You can connect an Excel file directly to your Power BI dataset.

Then, you can use **PivotTables, charts, and formulas** in Excel while the data stays live from Power BI Service.

🔮 Steps:

1. In Power BI Service → click **Dataset** → **Analyze in Excel**
2. It downloads a connection file (.odc)
3. Open in Excel → your Power BI data appears in PivotTable

- **Publish Excel to Power BI:**

Upload Excel models/reports directly to Power BI for interactive dashboards.

➔ *Best of both worlds: Excel for flexibility, Power BI for automation & visuals.*

2. Power BI + SQL Integration

Power BI connects natively to **SQL databases** (SQL Server, MySQL, PostgreSQL, etc.).

◆ Ways to Connect:

- **Import Mode:** Loads data from SQL into Power BI (faster visuals, but static).
- **DirectQuery Mode:** Connects live to SQL — updates instantly when the database changes.

✦ Steps:

1. In Power BI Desktop → **Get Data** → **SQL Server**
2. Enter **Server Name** and **Database Name**
3. Choose **Import** or **DirectQuery**
4. Select tables → Load → Done!

➔ *Use SQL integration for large, structured datasets and real-time reporting.*

3. Exporting Reports (PDF, PowerPoint, SharePoint)

Power BI lets you **export and share** your dashboards in multiple formats for presentations or documentation.

◆ PDF Export:

- Go to **File** → **Export** → **PDF**
- Exports all report pages as static visuals (good for offline use)

◆ PowerPoint Export:

- Go to **File** → **Export** → **PowerPoint**
- Each Power BI page becomes a PowerPoint slide (editable layout)

◆ SharePoint Integration:

- Embed Power BI reports directly in SharePoint Online
 - Reports stay **interactive** (unlike PDF)
- ➔ *Makes sharing insights easy — no need for users to open Power BI separately.*

4. Embedding Power BI in Apps/Sites

You can embed Power BI reports into **web apps, company portals, or external websites** to make analytics accessible anywhere.

◆ Options:

- **Power BI Embedded (for developers):**

- Use APIs or iframe code to integrate reports in web apps.
- Example: A company portal showing real-time sales dashboards.

- **Publish to Web (for public sharing):**

- Generates a shareable URL or embed code.
- ⚠ *Use only for non-sensitive data (it's public).*

- **Microsoft Teams Integration:**

- Add Power BI dashboards as tabs in Teams channels.

➔ *Embedding allows you to deliver insights directly where users work.*

⚡ Summary

Integration	What It Does	Example
Power BI + Excel	Analyze datasets in Excel (live connection)	PivotTables from Power BI data
Power BI + SQL	Connect directly to SQL databases	Real-time sales tracking
Exporting Reports	Share in PDF, PPT, SharePoint	Monthly management report
Embedding Reports	Show dashboards in apps/websites	Embed sales dashboard in portal

In Short:

Power BI becomes more powerful when connected:

Excel for analysis → SQL for data → SharePoint & Apps for sharing → Web embedding for accessibility.

10 Performance Optimization

When Power BI reports become **slow, heavy, or laggy**, performance optimization ensures smooth, quick dashboards — even with large datasets.

Let's break down each key concept 🙌

1. Reducing Dataset Size

A **smaller dataset = faster performance** ✅

Reducing dataset size helps Power BI load visuals quickly and consume less memory.

◆ Tips:

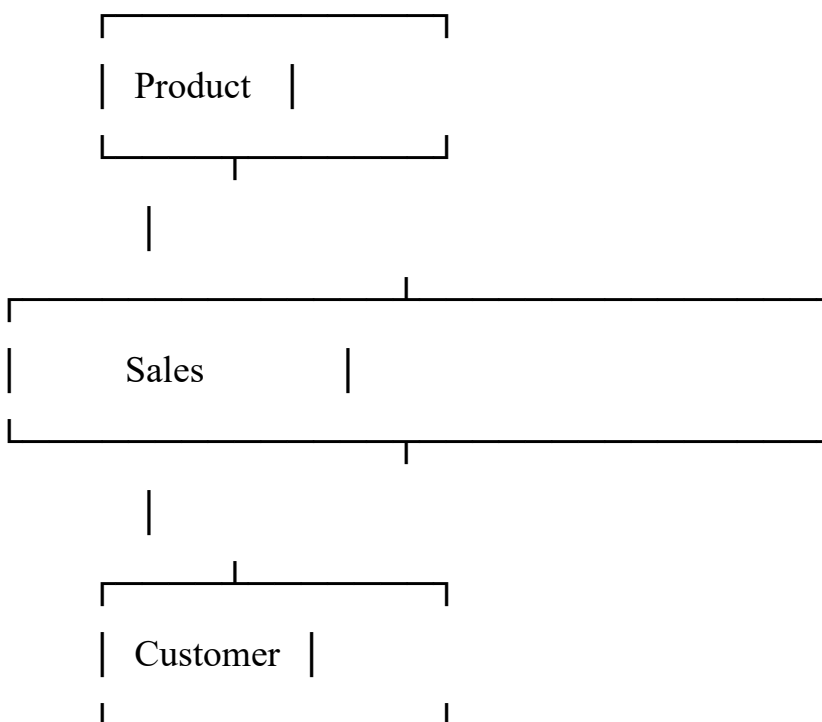
- **Remove unnecessary columns & tables**
→ Keep only what's used in visuals or calculations
 - **Filter data at import**
→ Example: Load only last 2 years instead of all history
 - **Use summary tables**
→ Aggregate at month/region level instead of daily details
 - **Use numeric data types wisely**
→ Avoid text when possible; compresses better
- ➔ *Smaller, cleaner datasets make dashboards lightning fast.*
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2. Optimizing Data Models (Star Schema Best Practices)

A **Star Schema** is the most efficient design for Power BI.

◆ **Structure:**

- **Fact Table:** contains numeric values (Sales, Profit, Quantity)
- **Dimension Tables:** contain descriptive data (Product, Customer, Region)



➔ *Avoid “Snowflake” schemas (too many linked dimensions); they slow down performance.*

3. Avoiding Bi-Directional Relationships (When Not Needed)

By default, relationships in Power BI flow **one way** (from dimension → fact). But when set to **bi-directional**, Power BI must evaluate filters both ways — which can slow down queries.

◆ **Best Practice:**

- Use **single-direction** relationships unless a visual needs both directions.

- Use **DAX functions** like CROSSFILTER() carefully if necessary.
 - ➔ *Minimize bi-directional filters — they're one of the biggest slowdowns in complex models.*
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4. Query Folding Concepts

Query Folding means Power BI pushes transformations back to the **data source** (like SQL) instead of processing them locally.

◆ **Example:**

When you filter or group data in Power Query, Power BI can translate that into an SQL command → so the database does the heavy work.

◆ **Best Practices:**

- Apply filters early (Power Query shows “Folding Enabled” in steps).
 - Keep folding active for performance (avoid custom M code that breaks folding).
- ➔ *Always let the source system do the heavy lifting when possible.*
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5. Performance Analyzer Tool

Power BI's **Performance Analyzer** helps you **find slow visuals or queries** in your report.

🔧 Steps:

1. Go to **View** → **Performance Analyzer**
2. Click **Start Recording**
3. Interact with report visuals
4. Check timings for:

- **DAX Query time**
- **Visual display time**
- **Other delays**

➔ *Use this tool to pinpoint which visuals or measures are slowing your report.*

6. Best Practices for DAX Performance

DAX (Data Analysis Expressions) can get complex — and slow if not written efficiently.

◆ Tips:

- Avoid using **CALCULATE inside CALCULATE** (nested context = slow)
- Use **variables (VAR)** to store intermediate results
→ Reduces repeated computation
- Prefer **SUMX / FILTER** only when needed (row-by-row = slower)
- Avoid **ALL()** when not necessary (removes all filters and expands calculations)
- Pre-aggregate data in Power Query instead of DAX when possible

➔ *Efficient DAX = fast queries and smoother visuals.*

Summary Table

Optimization Area	What It Means	Best Practice
Dataset Size	Amount of data loaded	Remove unused columns & rows
Data Model	Structure of tables	Use Star Schema
Relationships	How tables connect	Avoid bi-directional filters

Optimization Area	What It Means	Best Practice
Query Folding	Let source handle processing	Keep folding enabled
Performance Analyzer	Tool to find slow visuals	Use it before publishing
DAX Optimization	Faster calculations	Use VARs, avoid nested CALCULATE

In Short:

“Fast Power BI = Smart Data Model + Clean Queries + Efficient DAX.”

- Keep your model simple
 - Let your database do the work
 - Write optimized DAX
 - Test performance before publishing
-

1 2 Bonus: Advanced Topics in Power BI

Once you’ve mastered the basics, Power BI’s **advanced features** help you perform **deeper insights, automation, and predictive analytics**.

Let’s explore each one 🙌

1. Paginated Reports (SSRS in Power BI)

Paginated reports are designed for **pixel-perfect, printable reports** — ideal for invoices, forms, and financial statements.

- Developed using **Power BI Report Builder** (similar to SSRS).
- Unlike dashboards, they show **detailed table-style data** that fits across pages (“page by page,” hence *paginated*).

- Best for:
 - Official documents
 - Printable reports (PDFs)
 - Long tabular data (e.g., 10,000+ rows)

➔ *Use Power BI for visual dashboards; use paginated reports for printable, formatted outputs.*

2. Power Automate Integration (Workflow Automation)

Power Automate connects Power BI with other apps to **automate tasks** and create **data-driven workflows**.

◆ Examples:

- Send an email when sales fall below a threshold
- Trigger a Teams message when a dashboard updates
- Refresh a dataset automatically after data upload

✦ Steps:

1. In a Power BI visual → Click **More Options** → **Automate**
2. Choose or create a **Power Automate Flow**

➔ *Turns insights into instant actions — no manual monitoring needed!*

3. AI Visuals (Key Influencers & Decomposition Tree)

Power BI includes **AI-powered visuals** that help you find insights automatically — no coding needed.

◆ Key Influencers Visual

- Shows *which factors influence a certain outcome*.
Example: “What drives high customer satisfaction?”
→ It might show “Fast delivery” or “Discounts” as top influencers.

◆ **Decomposition Tree Visual**

- Breaks down metrics step-by-step.
Example: Sales → by Region → by Product → by Salesperson.
- Lets you explore data hierarchically and discover what drives performance.

➔ *AI visuals help find hidden patterns and causes behind trends.*

4. Python & R Integration with Power BI

Power BI allows you to use **Python** and **R scripts** for advanced analytics, visualizations, and data transformations.

◆ **In Power Query:**

You can run Python/R scripts to clean or process data before loading it.

◆ **In Visuals:**

Use **Python Visual** or **R Visual** to create:

- Advanced statistical charts (box plots, heatmaps, regression plots)
- Machine learning models (clustering, prediction)

✨ Steps:

1. Install Python/R on your system
2. Enable in **Options** → **Python/R scripting**
3. Add script visuals or transformations

➔ *Perfect for data scientists or analysts needing deeper analytics beyond built-in visuals.*

5. Advanced Analytics (Forecasting, Clustering, etc.)

Power BI includes **built-in analytics tools** to identify trends, patterns, and future predictions.

◆ Forecasting

- Predict future trends using existing time series data
- Add directly to line charts via **Analytics Pane** → **Forecast**

◆ Clustering

- Automatically groups similar data points (e.g., similar customers or regions)
- Available through **Scatter charts** or **AI Insights**

◆ Trend lines, outliers, moving averages

- Help explain performance and anomalies visually.

➔ *Advanced analytics transforms static dashboards into predictive, smart tools.*

Summary Table

Advanced Feature	What It Does	Example
Paginated Reports	Page-perfect printable reports	Invoice, monthly bill reports
Power Automate	Automate workflows	Email alert when sales drop
AI Visuals	Auto-insight generation	Find factors affecting profit
Python & R Integration	Custom scripts & visuals	Predict churn or cluster customers
Advanced Analytics	Forecasting & trend detection	Predict next month's revenue

In Short:

Power BI isn't just for dashboards anymore —

It's an **intelligent analytics platform** with automation, AI, and predictive power. 🚀

-  *Paginated reports* → for detailed printables
-  *Power Automate* → for instant actions
-  *AI visuals* → for smart insights
-  *Python/R* → for advanced analysis
-  *Forecasting & clustering* → for data science-level predictions

